

Atividade 1

Software Architectures

In modern software development, choosing the right architecture is essential. Systems ____ (must / can to / ought) be designed according to scalability, performance, and maintainability requirements.

One of the most traditional models is the monolithic architecture, where all components are part of a single system. As the application grows, developers ____ (are facing / is facing / was facing) increasing difficulty in maintaining the codebase.

In contrast, microservices architecture breaks the system into smaller services. Each service ____ (can / must to / ought) operate independently and communicate through APIs. While teams ____ (are working / worked / were working) on different services, coordination becomes more complex.

Another important model is the layered architecture, where the system is divided into layers. Each layer ____ (must / may to / ought) interact only with adjacent layers to maintain structure and organization.

The client-server architecture is widely used in web applications. In this model, clients request services while servers ____ (are processing / were processed / is process) those requests. At the same time, multiple clients ____ (are connecting / connected / connects) to the same server.

A different approach is the peer-to-peer (P2P) architecture, where each node ____ (can / can to / ought) act as both client and server. While data ____ (is being shared / shared / was being shared) across nodes, the system becomes more decentralized.

The event-driven architecture is used in systems that react to events. Components ____ (may / may to / ought) communicate through event streams. While events ____ (are being processed / were being processed / process), the system can respond in real time.

In distributed architecture, components are spread across multiple machines. Engineers ____ (must / must to / ought) ensure communication between nodes is reliable. While the system ____ (is running / was running / are running) across different environments, failures may occur in some parts without affecting the whole system.

During a system failure last week, several services ____ (were running / ran / are running) while engineers were trying to fix the issue. At that moment, users ____ (were experiencing / was experiencing / are experiencing) delays in the application.

Finally, in serverless architecture, developers do not manage servers directly. While the system ____ (is executing / executed / was executing) functions in response to events, cloud providers handle the infrastructure.

In conclusion, different architectures ____ (should / should to / ought) be selected based on the system's needs. While organizations ____ (are evolving / evolved / were evolving), their architectural choices also change over time.